Human-in-the-Loop Anomaly Detection for Journal Entry Testing

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Abstract

This exposé investigates the effectiveness of **Human-in-the-Loop (HITL)** feedback in improving anomaly detection for financial journal entries. The integration of machine learning models, particularly **Isolation Forest**, and human feedback reduces false positives, improving both the efficiency and accuracy of financial audits.

1 Introduction

In the world of financial audits, detecting anomalies in journal entries is a critical task for identifying errors or fraudulent activities. However, traditional methods often produce high rates of false positives, which require manual intervention. The solution? A cutting-edge integration of machine learning and human feedback: **Human-in-the-Loop Anomaly Detection**.

As noted by (1), anomaly detection has evolved from rule-based methods to data-driven techniques such as **Isolation Forest**, which is more effective at identifying outliers. This move to machine learning models has the potential to reduce the manual work required by auditors, but false positives remain a significant issue. Moreover, (3) emphasizes that **Human-in-the-Loop** methods are gaining momentum in machine learning for their ability to improve model performance iteratively.

2 Why It Matters

Current auditing methods are slow and inefficient. Auditors spend valuable time manually reviewing false alarms triggered by traditional rule-based systems. By automating anomaly detection and allowing auditors to provide feedback, we can significantly reduce errors, improve detection accuracy, and make audits more efficient.

Real-World Problem:

- Traditional rule-based systems = high false positives
- Auditors spend too much time on manual checks

The Goal:

- Reduce false positives
- Improve audit efficiency
- Enhance financial transparency

3 Objectives

This project aims to create a prototype system that integrates anomaly detection with **Human-in-the-Loop feedback**. The main objectives are:

1. **Anomaly Detection**: Use machine learning (Isolation Forest) to detect anomalies in journal entries.
2. **Human-in-the-Loop Feedback**: Auditors will label flagged anomalies, providing feedback to fine-tune the model.
3. **Metrics and Evaluation**: Track performance (Precision, Recall) before and after feedback integration, and visualize it in a user-friendly dashboard.
4. **Dashboard**: Display metrics and review queues, allowing auditors to quickly evaluate flagged entries.

4 Methodology

4.1 Step 1: Data Input

We will use structured journal entry data, consisting of financial transactions, with known anomalies (fraudulent or erroneous entries).

4.2 Step 2: Anomaly Detection (Machine Learning Model)

An **Isolation Forest** model will be used to detect unusual entries that differ from the norm, flagging potential errors or fraud. As stated in (2), **Isolation Forest** is highly effective in detecting outliers due to its ability to isolate anomalies rather than just distinguishing them from normal data.

4.3 Step 3: Human-in-the-Loop Integration

Feedback Mechanism: Auditors will review flagged anomalies and provide feedback (True Positive or False Positive).
Model Refinement: Feedback will be used to adjust thresholds or refine the model, improving accuracy.

As noted by (3), Human-in-the-Loop is crucial for maintaining a continuous feedback loop, where the model can adapt and improve based on real-time inputs from auditors, which helps to mitigate issues like **false positives**.

4.4 Step 4: Metrics & Dashboard

The dashboard will visually display important metrics to track the performance of the anomaly detection system. The dashboard will show:

- **Precision and Recall**: Measures of how well the system detects true positives while minimizing false alarms.
- **False Positive Rate**: Measures of how many flagged anomalies are actually false positives.
- **Review Queue**: A real-time view of anomalies flagged for auditor review, allowing auditors to act quickly and efficiently.

This real-time dashboard will allow auditors to actively monitor and manage flagged anomalies and adjust the model as necessary based on the feedback received.

4.5 Step 5: Evaluation

The system's success will be measured by comparing performance (Precision, Recall) before and after feedback integration, assessing how much the model improves over time.

5 System Model

Here's a simple flow of how the system works:

`Data Input` → `Anomaly Detection` → `Human-in-the-Loop Feedback` → `Model Adjustment` → `Metrics and Dashboard`

Explanation:

- Data Input: Journal entries (structured data)
- Anomaly Detection: Using Isolation Forest to detect outliers
- Human-in-the-Loop: Auditors review and provide feedback on flagged entries
- Model Adjustment: Feedback is used to adjust the detection model
- Metrics and Dashboard: Track the performance metrics and show flagged entries in a review queue

6 Evaluation Metrics

We will use the following metrics to evaluate the performance of the anomaly detection system:

- **Precision**: How many flagged anomalies are truly fraudulent or erroneous?
- **Recall**: How many actual anomalies are detected by the model?
- **False Positive Rate**: How many false alarms are triggered by the model?
- **Review Queue**: Number of flagged entries awaiting human review and feedback.

7 Conclusion

By combining machine learning with **Human-in-the-Loop feedback**, this project aims to dramatically improve the accuracy and efficiency of financial audits. Reducing false positives will lead to less manual review, saving auditors time and resources. Ultimately, this will improve the reliability and transparency of financial reporting, making audits more efficient and effective.

8 References

References

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